

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
 Department of Computer Science and Engineering
ASSESSMENT TOOL 1 – ANALYTICAL PROBLEM SOLVING

Course Code:	CSA0620	Course Title:	Design and Analysis of Algorithms
CO Assessed:	CO2 – Manipulation (BL1, BL2, BL3)	Assessment:	Assessment Tool 1 – Analytical Problem Solving
Weightage:	40% of CIA	Total Marks:	100
CO2 Statement:	Apply Brute Force technique to solve sorting, searching, Closest-Pair and Convex-Hull problems (BL1, BL2, BL3)		
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Section / Batch:	CSE AI	Date:	27-07-2026

Instructions to Students

- Answer the following question. Total Marks: 100.
- Analyze each problem carefully before applying the appropriate Brute Force technique.
- Clearly show all intermediate steps, comparisons, iterations, and logical reasoning used in solving the problems.
- Apply suitable algorithms for sorting, searching, Closest-Pair, and Convex-Hull problems wherever required.
- Justify the correctness and efficiency of the proposed solution using appropriate complexity analysis.
- Use diagrams, tables, or stepwise illustrations wherever necessary for better explanation.
- Maintain neat presentation, proper algorithmic notation, and structured analytical reasoning throughout the answers.

Question Type	Focus Area	Questions
Application-Based Questions	Apply Brute Force techniques for sorting, searching, Closest-Pair and Convex-Hull problem analysis	Q1

SECTION A – APPLICATION BASED QUESTIONS

Q1. Linear Search – Target Appearing Multiple Times

Given the unsorted dataset [23, 45, 23, 67, 89, 23, 12], apply Linear Search to find the first occurrence of the element 23. Clearly interpret the problem and explain the sequential search process step-by-step. Count the number of comparisons required before locating the first match. Analyze the best-case, average-case, and worst-case time complexity. Verify the correctness of the result. Interpret the efficiency of Linear Search when the target appears multiple times.

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

RUBRICS FOR CALCULATION

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%	Clearly understands the problem and accurately identifies all requirements	Good understanding with minor gaps	Basic understanding; some requirements unclear	Poor understanding or misinterpretation of the problem
Application of Concepts	30%	Accurately applies appropriate concepts to solve complex problems	Applies relevant concepts correctly with minor errors	Applies basic concepts with limited accuracy	Fails to apply appropriate concepts
Problem-Solving Approach	20%	Logical, well-structured, and efficient approach	Mostly logical approach with minor gaps	Basic approach with some inconsistencies	Illogical or unclear approach
Accuracy of Solution	20%	Completely correct solution with proper justification	Mostly correct with minor errors	Partially correct solution	Incorrect or incomplete solution
Clarity	10%	Clear, well-organized, and properly explained solution	Understandable with minor clarity issues	Limited clarity and explanation	Poorly presented and difficult to understand

Marks Evaluation:

Criteria	Wt.	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Problem Context	20%				
Application of Concepts	30%				
Problem-Solving Approach	20%				
Accuracy of Solution	20%				
Clarity	10%				
Total Marks (100):					

Faculty In-charge	Course Coordinator	Head of Department
Name: _____	Name: _____	Name: _____
Signature: _____	Signature: _____	Signature: _____
Date: _____	Date: _____	Date: _____

Given :

Dataset: [23, 45, 23, 67, 89, 23, 12]

Dataset Size (N): 7

Target Element: 23

Goal: Apply the linear search algorithm sequentially from left to right to find the first occurrence of the element 23 and return its index.

Algorithm for Linear Search:

1. Start: Begin at the first element (index 0) of the data-set.
2. Compare: Current element with target key.
3. Match check:
 - If $\text{array}[i] == \text{key}$, return the current index i (Target found).
4. Advance: if $\text{array}[i] \neq \text{key}$, increment index i to move to the next element.
5. Repeat: Steps 2 through 4 until either a match is found or all elements have been examined.
6. Terminate: If the loop finishes without finding key, return -1 (indicating the element is not present in the dataset).

Search Process:

- Step 1 (Index 0):
 - Compare element at index 0 (23) with key (23).
 - Result $23 == 23 \rightarrow$ Match Found
 - Action: Terminate ~~the~~ search and return index

Number of comparisons required:

- Total comparisons: 1 comparison is needed to locate the first occurrence.

Time Complexity Analysis:

- Best-case Complexity: $O(1)$
 - Condition: Occurs when the target element is at the very first position (index 0)
 - In this scenario: Only 1 comparison was performed, achieving the best-case time complexity.
- Average-Case Complexity: $O(N)$
 - Condition: On average, for uniformly distributed elements or multiple occurrences, the target is located in the middle region of the array.

Asymptotically, it runs in $O(N)$ time.
- Worst-Case Complexity: $O(N)$
 - Condition: If the target is at the last position or not present in the array at all, requiring N comparisons (7 comparisons for this dataset).

Efficiency Analysis:

- Impact of Duplicates: When a target element appears multiple times (at indices 0, 2 and 5 in this dataset), the likelihood of finding the target earlier in the search increases.
- Easy Termination (In this case): Standard linear searches stop when the first occurrence of the target element matches. So, we are at an advantage.